Customer Churn Analysis

A Data Analytics Project

Introduction: Project Goals

Understand Behavior

Examine customer actions and patterns.

Identify Drivers

Pinpoint key factors leading to churn.

Actionable Insights

Develop strategies for customer retention.

Data

Understand the current situation

Insight

Make sense of data and highlight where to target action

Action

Move from data-driven insight to targeted action to deliver impactful change

Workflow Overview

Our analysis is consolidated within a single notebook: my_script.ipynb.

01
Data Understanding

02
Data Cleaning

03
Exploratory Analysis

04
Customer Segmentation

05
Advanced Analysis

Data Understanding & Cleaning

Initial Assessment

- Checked data structure
- Identified missing values
- Verified data types

Refinement Process

- Cleaned inconsistencies
- Standardized formats
- Validated dataset integrity

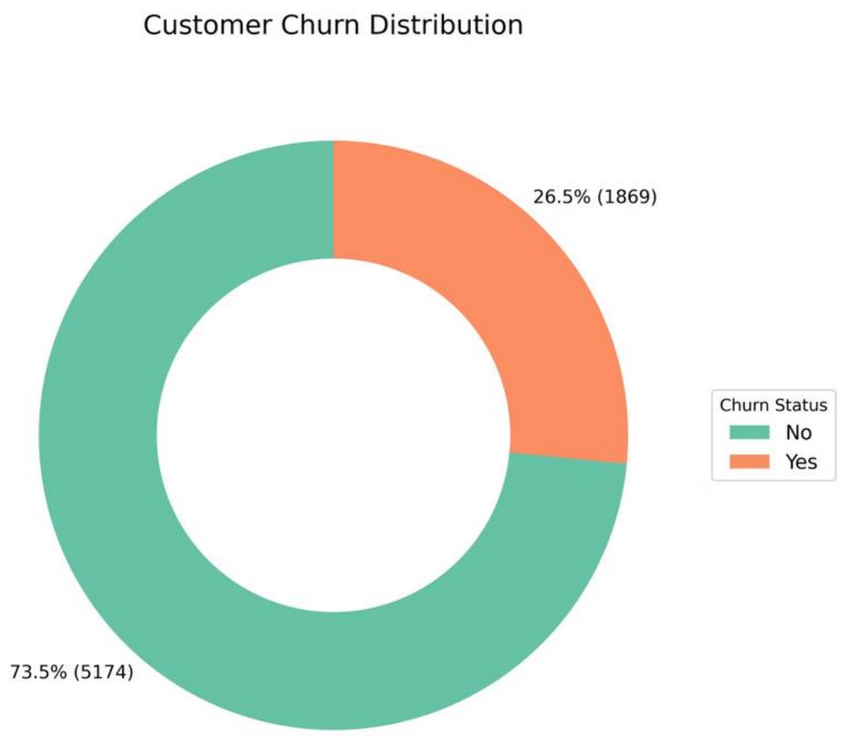
Exploratory Data Analysis (EDA)

Key Metrics

- Customer Demographics: Gender, age groups, senior citizen status
- Tenure Insights: Distribution of customer lifecycles (short, medium, long-term)
- Revenue Drivers: Monthly and total charges trends
- Churn Profile: Churn rate across segments and service types

Churn Balance

Assessed the balance of churned vs. non-churned customers.



Customer Segmentation

Customers were segmented into tenure buckets to analyze spending patterns.

Short-Term

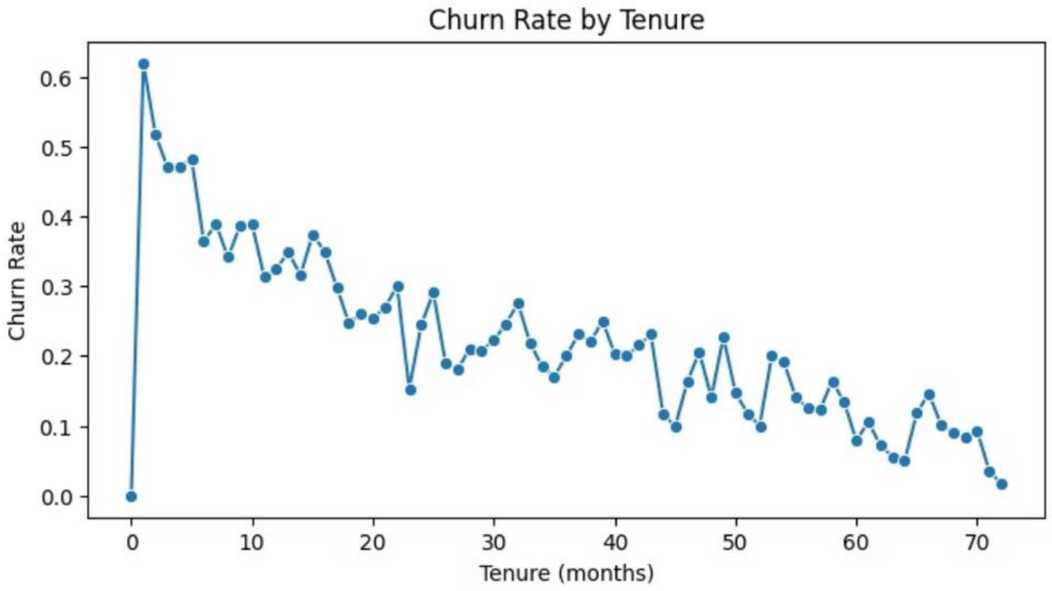
0–12 months

Mid-Term

13–36 months

Long-Term

37+ months



Advanced Analysis: Demographics

Churn rates were analyzed across various demographic and service-related factors.

- Customer Demographics: Gender, age groups, senior citizen status
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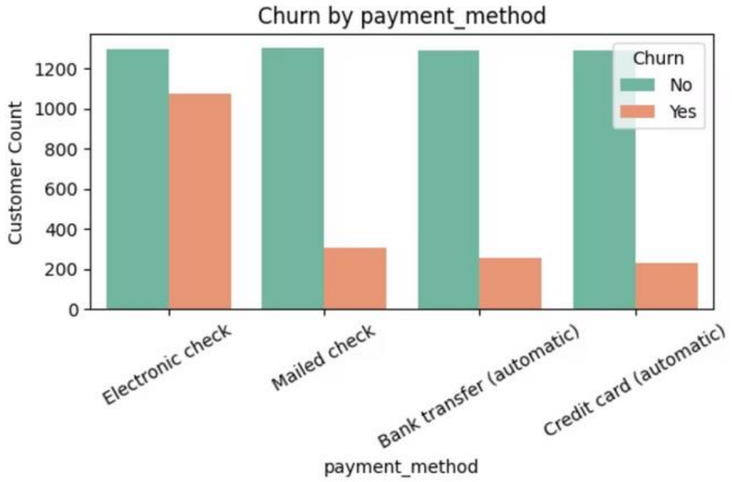
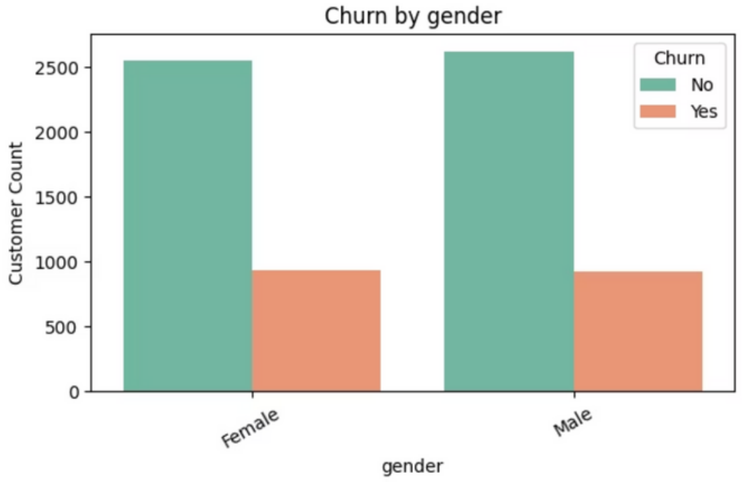
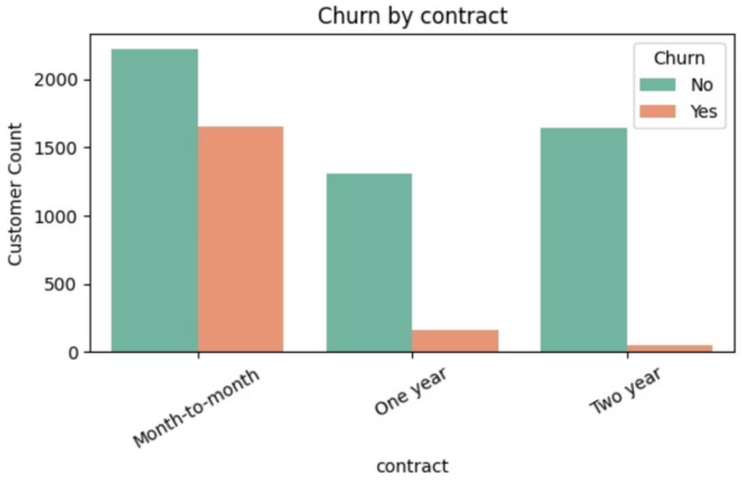




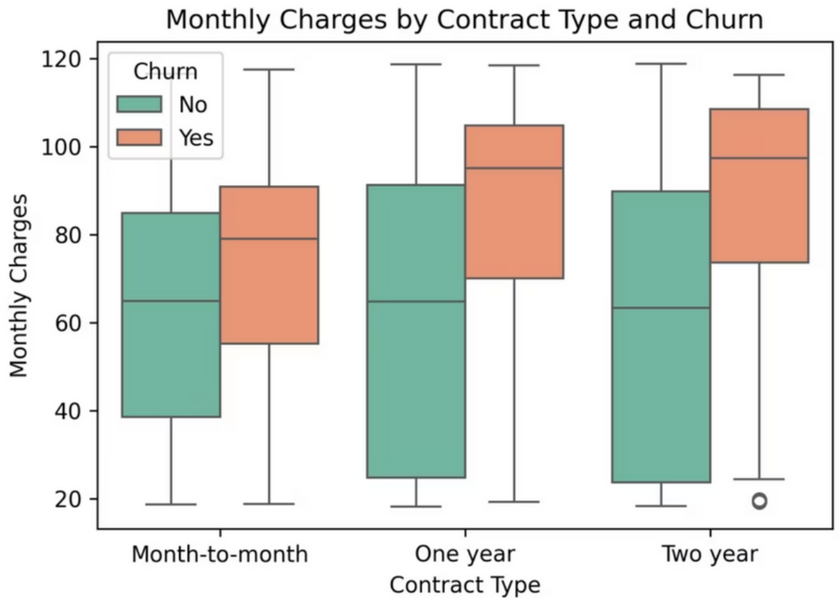


Advanced Analysis: Spending & Contracts

Contract types and monthly charges significantly impact churn.

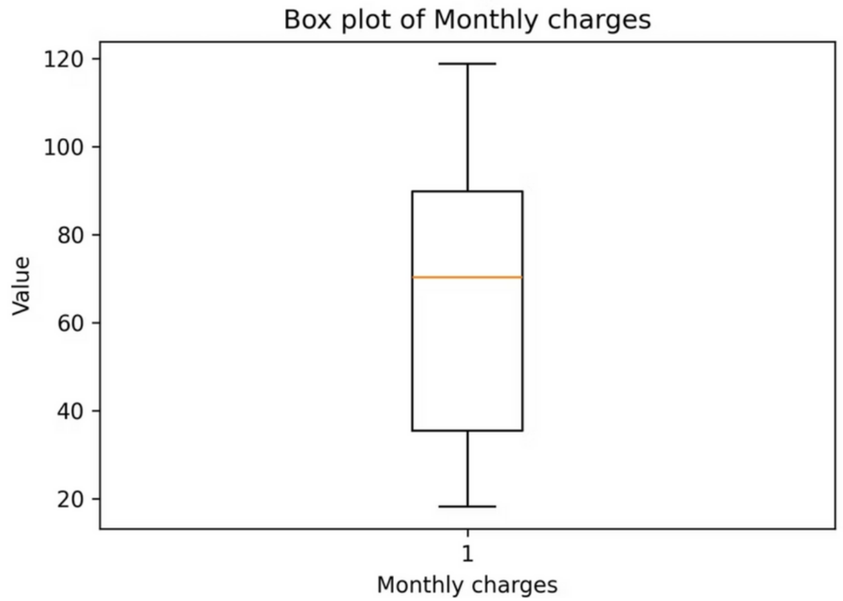
Contract Influence

- Compared churn across different contract types.
- Identified contracts with higher churn rates.



Spending Sensitivity

- Analyzed churn based on monthly charges.
- Boxplots illustrate charge distribution vs. churn.



Customer Churn Insights

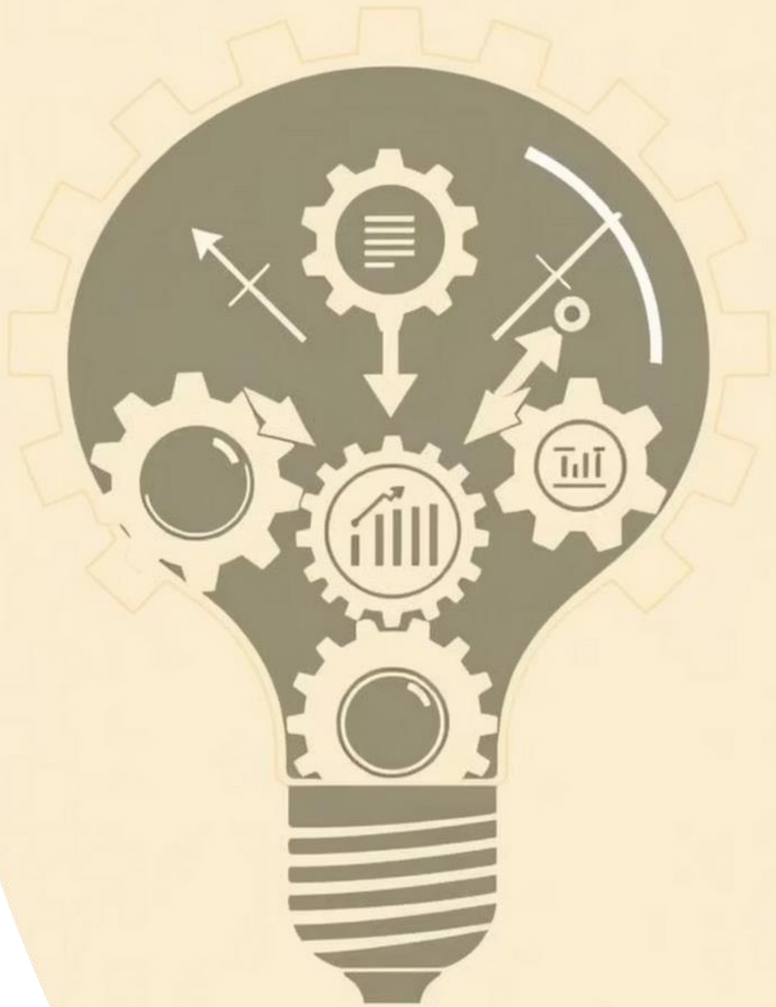
Leveraging data analytics to uncover critical customer behaviors, quantify retention performance, and formulate targeted strategies to reduce churn.

Customer Base Overview

- **Churn Rate:** 26.5% overall (1,869 churned out of 7,043 customers).
- **Revenue Distribution:** Majority of customers contribute low revenue, while a small high-value segment drives disproportionate revenue impact.
- **Customer Support:** Slow response times and unresolved issues.

Key Churn Patterns

- **New customer vulnerability:** 40–60% churn in first 10 months, stabilizing after 20+ months
- **Service gaps:** Customers lacking device protection show **78% higher churn rates**
- **Technology impact:** Fiber customers demonstrate elevated churn compared to DSL users

Revenue Insights

- Monthly charges range **$18–120**; median **$65–70**.
- Lifetime revenue is **extremely right-skewed** (majority low, few high).
- Top 1% customers generate **~$8,000+ total charges**, crucial to overall revenue.